



Technical Service Bulletin

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Camshaft Idler Plate or Camshaft Idler Gear Cover Seal Lubricating Oil Leak Misdiagnosed as Leaking Front Gear Housing Gasket and New SRTs

Camshaft Idler Plate or Camshaft Idler Gear Cover Seal Lubricating Oil Leak Misdiagnosed as Leaking Front Gear Housing Gasket and New SRTs

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- ISX15 CM2250 SN
- X15 CM2350 X114B
- X15 CM2350 X116B
- X15 CM2450 X124B
- X15 CM2450 X134B

Issue

Symptom:

- Lubricating oil leak noticed around front of the engine either at gear housing to cylinder block interface or lubricating oil pan to cylinder block interface.

Root Cause:

- Camshaft idler plate and/or adjustable camshaft idler gear cover seal lubricating oil leak.

Verification

- Clean the area around the front gear housing to cylinder block interface and add fluorescent tracer dye, Part Number 3376891, with leak test kit, Part Number 2892320, to aid in determining source of lubricating oil leak.
- Inspect the area between the cylinder head and front gear housing below the camshaft cover plate and adjustable camshaft idler cover plate (see Figure 1) and above the front gear housing to cylinder block joint.
- Lubricating oil leaking from either of these cover plates can travel down the side of the engine and the front gear housing to cylinder block joint which appears to be a gear housing

gasket leak (see yellow arrows in Figure 1, and Figure 2).

- Evidence of lubricating oil leak above the top of the front gear housing to cylinder block joint indicates the front engine lubricating oil leak is likely **not** the gear housing gasket, but the camshaft cover seal or adjustable camshaft idler cover plate seal.

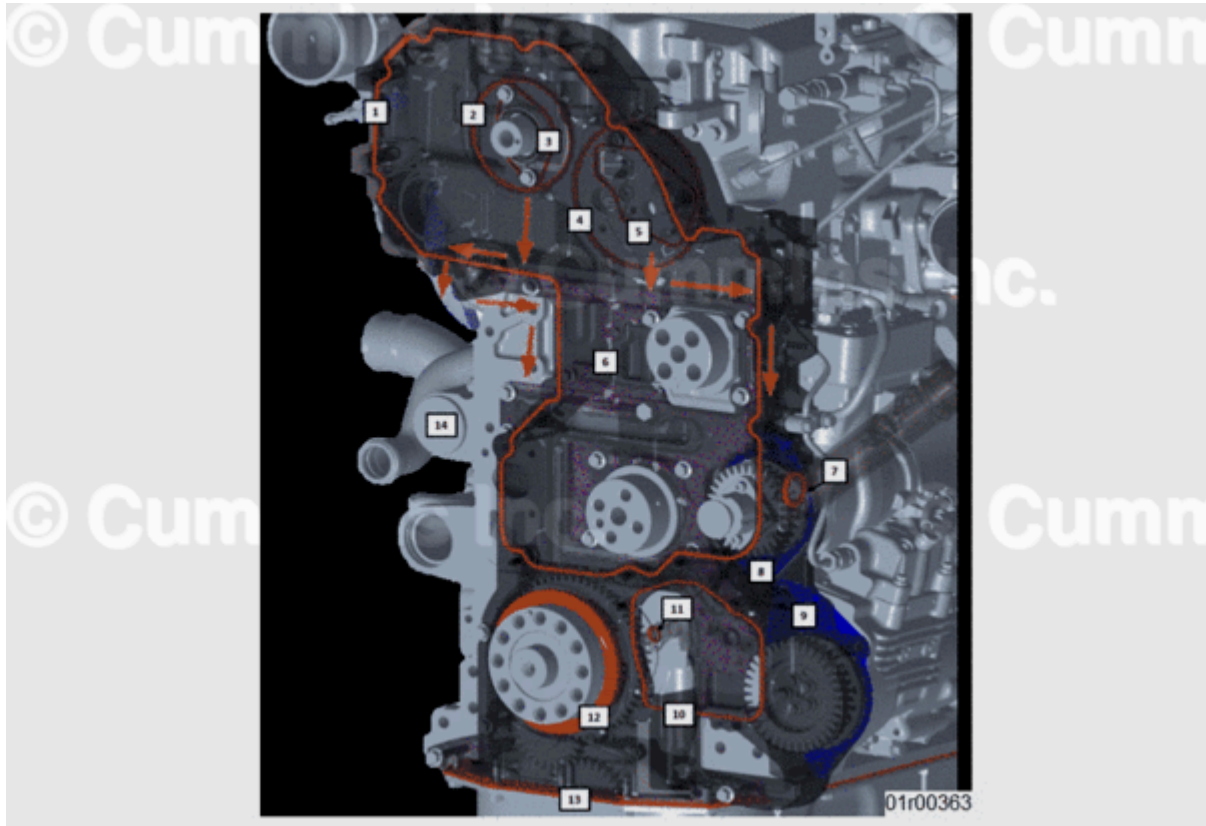


Figure 1, Camshaft Cover Plate Seal (3) And Adjustable Camshaft Idler Cover Plate Seal (5) And Typical Lubricating Oil Leak Flow from Seals.



Figure 2, Camshaft or Adjustable Camshaft Idler Gear Cover Seal Lubricating Oil Leak. View: Front of Exhaust Manifold with Accessories Removed.

Resolution

Complete the repair as normal. See corresponding Service Manual. Reference Procedure 002-050 in Section 2 or 001-036 in Section 1.

Description of Change

This document is to highlight that a lubricating oil leak from the camshaft cover seal or adjustable camshaft idler cover plate seal can appear to be a lubricating oil leak from the front gear housing gasket. New SRTs have been released for camshaft cover plate seal and adjustable camshaft idler cover plate seal repairs. See Table 1 below.

Table 1, New SRTs

Service Model Name	SRT Name	SRT Number
X15 CM2350 X114B	Camshaft Retainer Plate – Remove and Install	02-0HA
	Idler Gear, Adjustable – Remove and Install	01-2D4
X15 CM2350 X116B	Camshaft Retainer Plate – Remove and Install	02-0HB
	Idler Gear, Adjustable – Remove and Install	01-2D5

Table 1, New SRTs		
Service Model Name	SRT Name	SRT Number
X15 CM2450 X124B	Camshaft Retainer Plate – Remove and Install	02-0HC
	Idler Gear, Adjustable – Remove and Install	01-2D6
X15 CM2450 X134B	Camshaft Retainer Plate – Remove and Install	02-0HD
	Idler Gear, Adjustable – Remove and Install	01-2D7

Document History

Date	Details
2021-4-16	Module Created
2022-8-24	Added ISX15 CM2250 SN.

Last Modified: 24-Aug-2022
